

Logical & Verbal Reasoning — Complete Preparation Guide

For CSE/SDE Placement Tests (TCS NQT, Infosys, Wipro, Capgemini, etc.)

Study Notes

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1. Number and Letter Series

1.1 Key Concepts

- Look for the pattern: arithmetic (+/- constant), geometric ($\times/\div$ constant), squares/cubes, alternating patterns, or combination of two interleaved series.
- **Letter series** map letters to numbers (A=1...Z=26) and apply the same numeric logic.
- Common patterns: difference of differences (2nd order AP), prime number series, Fibonacci-like (each term = sum of previous two).

1.2 Important Notes / Shortcuts

- Write out the differences between consecutive terms first — this reveals AP, GP, or a repeating pattern instantly.
- For letter series, convert to numbers only if a numeric pattern isn't obvious directly from letter jumps.
- Watch for "odd one out" series questions — find the term that breaks an otherwise consistent pattern.

1.3 Solved Examples

Example 1: Find the next term: 2, 6, 12, 20, 30, ?

Approach: Differences: 4, 6, 8, 10 (increasing by 2). Next difference = 12. Answer = $30 + 12 = 42$. (Pattern: $n(n+1)$)

Example 2: Find the missing letter: B, E, H, K, ?

Approach: Each letter jumps +3: B(2)→E(5)→H(8)→K(11)→**N(14)**.

Example 3: Find the odd one out: 3, 5, 11, 14, 17, 21

Approach: All except 14 are odd numbers → **14** is the odd one out.

1.4 Practice MCQs

Q1. Find the next term: 1, 4, 9, 16, 25, ? (a) 30 (b) 36 (c) 32 (d) 49

Q2. Find the next term: 3, 6, 11, 18, 27, ? (a) 36 (b) 38 (c) 40 (d) 34

Q3. Find the missing term: AZ, BY, CX, ? (a) DW (b) EW (c) DV (d) DX

Q4. Find the next term: 5, 11, 23, 47, ? (a) 93 (b) 95 (c) 96 (d) 94

Q5. Find the odd one out: 2, 3, 5, 9, 11, 13 (a) 2 (b) 5 (c) 9 (d) 13

Answer Key: 1-(b) n^2 pattern | 2-(c) diff 3, 5, 7, 9, 11 → $+11=38$ → recheck: $27+13=40$ (diffs 3, 5, 7, 9, 11 → next diff 11) → (c) 40 | 3-(a) 1st letter +1, 2nd letter -1 | 4-(b) $\times 2 + 1$ pattern: $5 \rightarrow 11 \rightarrow 23 \rightarrow 47 \rightarrow 95$ | 5-(c) 9 is not prime

2. Coding-Decoding

2.1 Key Concepts

- **Letter coding:** each letter is shifted by a fixed number (e.g., +1, +2) or replaced by a symmetric letter ($A \leftrightarrow Z$, $B \leftrightarrow Y$, i.e. reverse alphabet).
- **Number/symbol coding:** words are assigned numbers or symbols based on position or a substitution key.
- **Word coding:** entire words are substituted with other words based on a given key; decode by matching patterns across multiple given statements.

2.2 Important Notes / Shortcuts

- Write the alphabet with position numbers ($A=1 \dots Z=26$) alongside the reverse ($Z=1 \dots A=26$) — makes shift and mirror patterns instantly visible.
- In word-substitution coding puzzles, find the word/symbol common to two statements — it identifies that shared word's code directly.

2.3 Solved Examples

Example 1: If CAT is coded as DBU, how is DOG coded?

Approach: Each letter shifted +1: $C \rightarrow D$, $A \rightarrow B$, $T \rightarrow U$. Similarly $D \rightarrow E$, $O \rightarrow P$, $G \rightarrow H \rightarrow$ **EPH**.

Example 2: In a code, "roses are red" is "3 5 8", "lily are white" is "5 9 6", and "red is bright" is "3 7 2". What is the code for "red"?

Approach: "red" appears in statements 1 and 3, shared code between "3 5 8" and "3 7 2" is **3**.

Example 3: If in a certain language, MADRAS is coded as NBESBT, how is BOMBAY coded?

Approach: Every letter shifted +1: $M \rightarrow N$, $A \rightarrow B$, $D \rightarrow E \dots$. Similarly $B \rightarrow C$, $O \rightarrow P$, $M \rightarrow N$, $B \rightarrow C$, $A \rightarrow B$, $Y \rightarrow Z \rightarrow$ **CPNCBZ**.

2.4 Practice MCQs

Q1. If PENCIL is coded as QFODJM, what is the code for BOOK? (a) CPPL (b) CPOL (c) DPPL (d) CQPL

Q2. If FRIEND is written as HRKGPF (each letter +2), how is CANDLE written? (a) EBPFG (b) ECPFG (c) ECPFOG (d) EDPFG

Q3. In a code, TABLE is written as UBCMF. How is CHAIR written? (a) DIBJS (b) DIBJT (c) DJBIS (d) DIAJS

Answer Key: 1-(a) +1 shift | 2-(c) recompute: $C+2=E$, $A+2=C$, $N+2=P$, $D+2=F$, $L+2=N$, $E+2=G \rightarrow$ ECPFG $\rightarrow$ (b) | 3-(a) +1 shift each letter

3 3. Blood Relations

3.1 3.1 Key Concepts

- Draw a **family tree diagram** for every problem — never try to solve purely mentally.
- Use symbols: + for male, – for female, = for marriage, | for parent-child link.
- Key relation chains to memorize:
 - Father's/Mother's son = Brother
 - Father's/Mother's daughter = Sister
 - Father's father = Grandfather
 - Mother's brother = Maternal Uncle
 - Father's sister = Paternal Aunt
 - Son's wife = Daughter-in-law
 - Sister's husband = Brother-in-law

3.2 3.2 Important Notes / Shortcuts

- Start from the person mentioned last in the question and work backward through the tree.
- Gender is often not explicitly stated — infer it from relation words like “son,” “daughter,” “father,” “mother.”
- For “pointing to a photograph” questions, carefully track whose relative is being described (the speaker's or the person pointed at's).

3.3 3.3 Solved Examples

Example 1: Pointing to a man, a woman says, “His mother is the only daughter of my mother.” How is the woman related to the man?

Approach: “Only daughter of my mother” = the woman herself. So the man's mother = the woman → **Mother**.

Example 2: A is B's sister. C is B's mother. D is C's father. E is D's mother. How is A related to D?

Approach: A is B's sister → A is C's daughter. D is C's father → D is A's grandfather → A is D's **Granddaughter**.

Example 3: Introducing a man, a woman said, “He is the son of my grandfather's only son.” How is the woman related to the man?

Approach: “My grandfather's only son” = the woman's father (since he's an only son and grandfather's son). So the man is the woman's father's son → **Brother**.

3.4 3.4 Practice MCQs

Q1. Pointing to a photograph, a man says, “She is the daughter of my grandfather's only son.” How is the woman related to the man? (a) Sister (b) Mother (c) Aunt (d) Cousin

Q2. A is the father of B. C is the son of A. D is the brother of C. What is the relation between D and B? (a) Same person / Brother (b) Father (c) Uncle (d) Cannot be determined

Q3. P is Q's brother. R is Q's mother. S is R's father. T is S's mother. How is P related to S? (a) Grandson (b) Grandfather (c) Son (d) Father

Answer Key: 1-(a) grandfather's only son = the man's father, so she's his sister | 2-(a) C and D are both A's children like B → D is B's brother | 3-(a) P is Q's brother, Q is R's child, R is S's child → P is S's grandson

4 4. Direction Sense

4.1 4.1 Key Concepts

- Standard directions: N, S, E, W, and NE, NW, SE, SW.
- **Left turn** rotates you 90° counter-clockwise; **right turn** rotates 90° clockwise.
- Facing North: Left $\rightarrow$ West, Right $\rightarrow$ East. Facing South: Left $\rightarrow$ East, Right $\rightarrow$ West.
- Use a coordinate system: North = $+y$, South = $-y$, East = $+x$, West = $-x$. Track final (x,y) position.
- **Shortest distance** between start and end point = straight-line (Pythagorean) distance, not the path walked.

4.2 4.2 Important Notes / Shortcuts

- Always draw the path on paper/mentally as a grid — direction problems are almost always faster solved visually.
- At sunrise, shadow falls to the West; at sunset, shadow falls to the East (sun rises in the East).

4.3 4.3 Solved Examples

Example 1: A man walks 5 km North, then 3 km East, then 5 km South. How far is he from the starting point?

Approach: North and South cancel ($5-5=0$). Only East 3 km remains $\rightarrow$ **3 km**.

Example 2: Starting from a point, Rahul walks 10m North, turns right and walks 10m, turns right again and walks 10m. Find his distance and direction from the start.

Approach: North 10 $(0,10) \rightarrow$ right turn (facing East) walk 10 $\rightarrow (10,10) \rightarrow$ right turn (facing South) walk 10 $\rightarrow (10,0)$. Distance from start = $\sqrt{(10^2+0^2)} =$ **10m, to the East**.

4.4 4.4 Practice MCQs

Q1. A person walks 8 km South, then 6 km West. How far is he from the starting point? (a) 10 km (b) 14 km (c) 12 km (d) 8 km

Q2. Facing North, if you turn 90° clockwise, then 180° , which direction do you face? (a) North (b) South (c) West (d) East

Q3. Ravi walks 20m East, then turns left and walks 20m, then turns left again and walks 20m. What direction is he from the start, and how far? (a) 20m West (b) 20m East (c) 20m North (d) 40m East

Answer Key: 1-(a) $\sqrt{(8^2+6^2)}=10$ | 2-(c) $N \rightarrow (90^\circ CW) E \rightarrow (180^\circ) W$ | 3-(b) East20 $\rightarrow$ North20 $\rightarrow$ West20, net: East 0, North 20 $\rightarrow$ wait recompute: start(0,0) $\rightarrow$ East(20,0) $\rightarrow$ left(facing North)walk20 $\rightarrow$ (20,20) $\rightarrow$ left(facing West)walk20 $\rightarrow$ (0,20). Distance=20m North $\rightarrow$ (c)

5. Syllogisms

5.1 Key Concepts

- A syllogism has two or more **premises/statements** and a **conclusion** that must follow logically — regardless of whether it matches real-world fact.
- Standard statement types: “All A are B”, “No A is B”, “Some A are B”, “Some A are not B”.
- **Venn diagram method** is the most reliable: draw circles for each category and check which conclusions are forced true in every possible diagram.
- A conclusion is valid only if it holds in **every** possible Venn diagram consistent with the premises — not just one.

5.2 Important Notes / Shortcuts

- “All A are B” does NOT imply “All B are A” (not reversible). “Some A are B” DOES imply “Some B are A” (reversible).
- If two statements are both “Some” statements, generally no definite conclusion can be drawn (except in some possibility cases).
- Complementary pairs (either-or conclusions): when two conclusions together cover all possibilities, at least one must follow — a common trap.

5.3 Solved Examples

Example 1: Statements: All cats are animals. All animals are mammals. Conclusion: All cats are mammals.

Approach: Draw nested circles: Cats $\subset$ Animals $\subset$ Mammals. So Cats $\subset$ Mammals $\rightarrow$ Conclusion is **valid**.

Example 2: Statements: Some pens are books. Some books are tables. Conclusion: Some pens are tables.

Approach: The overlap between Pens-Books and Books-Tables need not intersect at all — multiple valid Venn diagrams exist where pens and tables don’t overlap. Conclusion **does not follow**.

Example 3: Statements: No door is a window. All windows are glass. Conclusions: (I) No door is glass. (II) Some glass is not a door.

Approach: Since windows (a subset of glass) are entirely excluded from doors, but glass may have other parts overlapping doors or not — (I) doesn’t necessarily follow, but since windows $\subset$ glass and no windows are doors, that portion of glass is definitely “not a door” $\rightarrow$ **(II) follows, (I) does not**.

5.4 Practice MCQs

Q1. Statements: All roses are flowers. Some flowers fade quickly. Conclusion: Some roses fade quickly. (a) Follows (b) Does not follow

Q2. Statements: All students are hardworking. No hardworking person is lazy. Conclusion: No student is lazy. (a) Follows (b) Does not follow

Q3. Statements: Some books are pens. All pens are red. Conclusions: (I) Some books are red. (II) All books are red. (a) Only I follows (b) Only II follows (c) Both follow (d) Neither follows

Answer Key: 1-(b) the “some flowers” that fade may not include roses | 2-(a) valid syllogism chain | 3-(a) only I is guaranteed

6. Seating Arrangement and Puzzles

6.1 Key Concepts

- **Linear arrangement:** people seated in a row facing one direction (or some facing north, others south — read carefully).
- **Circular arrangement:** people seated around a table, facing inward or outward — this flips left/right.
- **Puzzle grids:** use a table (rows = people, columns = attributes) to track multiple constraints (age, profession, city, etc.) simultaneously.

6.2 Important Notes / Shortcuts

- Start with the most restrictive clue (a fixed position, e.g., “X sits at the leftmost end”) and build outward.
- In circular arrangements facing **inward**, left/right is counter-clockwise/clockwise as normally understood; facing **outward** reverses it.
- Use elimination grids for puzzles with multiple categories — mark **X** for eliminated combinations and **✓** once confirmed.
- Re-read the question after solving — puzzles often ask about a specific derived relationship, not just “who sits where.”

6.3 Solved Example

Example: Five friends A, B, C, D, E sit in a row facing North. B sits second from the left. D sits immediately right of B. A sits at one of the ends. C sits immediately left of E. Find the arrangement.

Approach: Positions 1-5. B is at position 2. D is at position 3 (immediately right of B). A is at an end → position 1 or 5. Remaining people C and E must be placed with C immediately left of E — the only consecutive empty pair remaining is positions 4-5 (if A is at 1) or 1 not available if A takes 5, leaving no consecutive pair, so **A is at position 1**, C at 4, E at 5.

Final arrangement: **A, B, D, C, E** (positions 1 to 5).

6.4 Practice MCQ

Q1. Six people P, Q, R, S, T, U sit around a circular table facing the center. Q sits third to the right of P. R sits immediately left of Q. S sits immediately right of P. If T and U are the remaining two seats, and T sits immediately left of R, where does U sit relative to P?

- (a) Immediately left of P (b) Immediately right of S (c) Opposite P (d) Between S and Q

Answer: By elimination after placing P, Q(3rd right of P), R(left of Q), S(right of P), T(left of R) — the only remaining seat for U works out to be **(a) immediately left of P** (verify by drawing the circle with 6 positions and placing each constraint step by step).

Note: Seating/puzzle questions are best practiced by actually drawing grids on paper — reading a solved answer helps less than solving 15-20 of these yourself. Prioritize drawing the diagram over mental solving.

7 7. Analogies and Classification

7.1 7.1 Key Concepts

- **Analogy:** A is to B as C is to ? — identify the *relationship* between the first pair (e.g., part-whole, cause-effect, synonym, function) and apply it to the second.
- **Classification (odd one out):** find the item that doesn't share the common property/category of the others.
- Common relationship types: Synonym, Antonym, Part-Whole, Worker-Tool, Animal-Young one, Cause-Effect, Degree (small-medium-large).

7.2 7.2 Solved Examples

Example 1: Doctor : Hospital :: Teacher : ?

Approach: Relationship = person : workplace. Teacher works at → **School**.

Example 2: Find the odd one out: Mango, Apple, Potato, Banana

Approach: All are fruits except Potato, which is a vegetable → **Potato**.

Example 3: Pen : Write :: Knife : ?

Approach: Relationship = tool : function. Knife is used to → **Cut**.

7.3 7.3 Practice MCQs

Q1. Bird : Nest :: Man : ? (a) Cave (b) House (c) Hotel (d) Building

Q2. Find the odd one: Triangle, Square, Circle, Cube (a) Triangle (b) Square (c) Circle (d) Cube

Q3. Water : Thirst :: Food : ? (a) Hunger (b) Eat (c) Taste (d) Stomach

Answer Key: 1-(b) | 2-(d) Cube is 3D, others are 2D shapes | 3-(a)

8 8. Statement and Assumption / Conclusion / Course of Action

8.1 8.1 Key Concepts

- **Assumption:** something implicitly taken for granted for the statement to make sense — not stated, but necessary.
- **Conclusion:** something that logically and definitely follows from the statement (not an inference requiring outside knowledge).
- **Course of Action:** a step that is practical, follows directly from the problem described, and addresses it reasonably.
- **Cause and Effect:** determine which of two given statements is the cause and which is the effect, or if they're independent.

8.2 8.2 Important Notes / Shortcuts

- For assumptions: use the “negation test” — if negating the assumption makes the statement illogical/pointless, it IS a valid assumption.
- Conclusions must not require any external/outside knowledge — only what's stated or a very direct, certain implication.
- Avoid options that are too broad, moralizing, or impractical for “course of action” questions — the best action is specific and directly addresses the stated problem.

8.3 8.3 Solved Example

Example: Statement: “Please use the side entrance as the main gate is under repair.” Assumption: People will comply and use the side entrance if informed.

Approach: Negate: “People will NOT comply even if informed” → this makes putting up the notice pointless. So the assumption **is implicit and valid**.

8.4 8.4 Practice MCQ

Q1. Statement: “The company has decided to offer work-from-home twice a week to boost employee morale.” Assumption: Employees value flexibility in work location. (a) Assumption is implicit (b) Assumption is not implicit

Answer: 1-(a) — the policy is designed on the premise that flexibility matters to employees; negating it would make the decision pointless.

9. Clocks and Calendars

9.1 Key Concepts & Formulas

Clocks: - The minute hand moves 6° per minute; the hour hand moves 0.5° per minute. - Angle between hour and minute hands at H hours and M minutes: $|30H - 5.5M|$ degrees (take the smaller of this and 360 minus this). - Hands overlap (coincide) 11 times every 12 hours (not 12), roughly every $65 \frac{5}{11}$ minutes. - A clock is a “mirror image” of itself at certain times — mirror time = 11:60 – actual reading (in hh:mm).

Calendars: - **Odd days** determine the day of the week. A normal year has 1 odd day ($365 \bmod 7 = 1$); a leap year has 2 odd days. - Century odd days: 100 years $\rightarrow$ 5 odd days; 200 years $\rightarrow$ 3 odd days; 300 years $\rightarrow$ 1 odd day; 400 years $\rightarrow$ 0 odd days (400 is a special case, fully divisible cycle). - Odd days to weekday mapping: 0=Sunday, 1=Monday, 2=Tuesday, 3=Wednesday, 4=Thursday, 5=Friday, 6=Saturday. - Leap year rule: divisible by 4, but if divisible by 100 must also be divisible by 400 to remain a leap year.

9.2 Solved Examples

Example 1: Find the angle between the hour and minute hands at 3:40.

Approach: $|30 \times 3 - 5.5 \times 40| = |90 - 220| = 130^\circ$. Since $130 < 180$, angle = **130°** .

Example 2: What day of the week was 15th August 1947?

Approach: (Standard result, computed via odd-day method) — **Friday**.

Example 3: At what time between 4 and 5 o'clock will the hands of a clock coincide?

Approach: Hands coincide when $|30H - 5.5M| = 0$. At $H=4$: $120 = 5.5M \rightarrow M = 21 \frac{9}{11}$. So time = **4:21 $\frac{9}{11}$** .

9.3 Practice MCQs

Q1. Find the angle between the hands of a clock at 6:30. (a) 0° (b) 15° (c) 30° (d) 45°

Q2. If 1st January 2023 was a Sunday, what day was 1st January 2024? (a) Sunday (b) Monday (c) Tuesday (d) Wednesday

Q3. How many times do the hands of a clock coincide in a day (24 hours)? (a) 20 (b) 22 (c) 24 (d) 44

Answer Key: 1-(b) $|30 \times 6 - 5.5 \times 30| = |180 - 165| = 15^\circ$ | 2-(b) 2023 non-leap, 1 odd day $\rightarrow$ Monday | 3-(b) 11 times per 12 hours $\times 2 = 22$

10 Quick Reference Sheet

Topic	Key Trick
Series	Check 1st and 2nd order differences first
Coding	Map A=1...Z=26, look for constant shift
Blood relations	Always draw the family tree
Direction	Use x-y coordinate tracking
Syllogism	Venn diagram must hold in every case
Seating	Start from the most restrictive clue
Clock angle	$ 30H - 5.5M $
Calendar	Normal year = 1 odd day, Leap year = 2 odd days

End of Reasoning Guide. Reasoning improves fastest through volume — solve 20-25 puzzles/seating arrangement sets under time pressure (under 5 min per set) before your test.